

1 Some f -divergences

Here are four notions of distance, between two probability distributions P and Q . Define the likelihood ratio $L(X) = \frac{q(X)}{p(X)}$ using the respective densities.

- The L_1 **distance**, or **Total Variation distance**, is given by

$$V(P, Q) := \frac{1}{2} \mathbb{E}_P[|1 - L(X)|] = \frac{1}{2} \|P - Q\|_1.$$

- The **Hellinger distance** is given by

$$H^2(P, Q) := 1 - \mathbb{E}_p \sqrt{L(X)} = \frac{1}{2} \mathbb{E}_p \left[\left(1 - \sqrt{L(X)} \right)^2 \right].$$

- The **Bhattacharyya distance** is given by

$$D_B(P, Q) := -\ln \mathbb{E}_P \sqrt{L(X)} = -\ln (1 - H^2(P, Q)).$$

- The **Kullback-Leibler divergence** is given by

$$D_{KL}(P \| Q) := -\mathbb{E}_P \ln L(X).$$

Now let us relate the 4 distances as follows.

First, the total variation distance squared is less than the Hellinger distance. Notice that by the Cauchy Schwarz inequality,

$$\begin{aligned} L_1^2(P, Q) &\leq \frac{1}{4} \cdot \int (\sqrt{p} + \sqrt{q})^2 dP \cdot \int (\sqrt{p} - \sqrt{q})^2 dP \\ &= \frac{1}{4} \cdot \int p + 2\sqrt{pq} + q dP \cdot 2H^2(P, Q) \\ &= \frac{1}{4} \cdot (2(1 - H^2(P, Q)) + 1 + 1) \cdot 2H^2(P, Q) \\ &= (2 - H^2(P, Q))H^2(P, Q) \\ &\leq 2H^2(P, Q) \end{aligned}$$

Next, the Hellinger distance is bounded by the Bhattacharyya distance. To see this, note that $\log x \leq x - 1$ by standard concavity argument and their meeting at

$x = 1$. By writing $D_B(P, Q) = -\log \int \sqrt{qp} \, dP$ and $H^2(P, Q) = 1 - \int \sqrt{qp} \, dP$ and then rearranging, we get

$$D_B(P, Q) \geq H^2(P, Q).$$

Finally, the Bhattacharyya distance is bounded by the KL divergence. Noting that $D_B(P, Q) = -\log \mathbb{E}_P \sqrt{q/p}$ and $KL(P, Q) = -\mathbb{E}_P \log(q/p)$, Jensen's inequality gives

$$\begin{aligned} \mathbb{E}_P \log \sqrt{q/p} &\leq \log \mathbb{E}_P \sqrt{q/p} \\ \iff KL(P, Q) &\geq 2D_B(P, Q). \end{aligned}$$